

The Goodwins' Heirloom Artist Pencil Case

Custom Two-Tone Woodworking Build Plan for **Debbie Goodwin** (2-Time State Champion Artist)



🎨 Overview & Two-Tone Design Concept

Designed specifically for **Prismacolor Premier** and **Faber-Castell Polychromos** colored pencils.

Two-Tone Craftsmanship: Main box & 4 trays are crafted from **Rough-Cut Black Walnut**, while the 8 cantilever linkage arms are milled from **Rough-Cut White Ash** scraps. The pale, creamy grain of White Ash against dark chocolate Walnut creates a striking studio contrast, accentuating the mechanical accordion movement and providing superior pivot hole durability (Ash Janka hardness ~1,320 lbf).

🎨 Master Finisher Specification:

- **Interior Trays & Lid Undersides: Danish Oil** (Deep grain penetration, zero-tack finish that won't stick to delicate pencil wax/oil cores).
- **Exterior Surfaces & White Ash Arms: Pure Tung Oil** (Superior UV resistance, rich amber warmth, hard curing coat for travel protection).

⚖️ Weight & Ergonomics Breakdown:

- **Empty Case Weight:** ~7.5 lbs (~3.4 kg) (*Walnut Body ~6.7 lbs + White Ash Arms ~0.3 lbs + Hardware ~0.5 lbs*)
- **Fully Loaded Weight (120+ Pencils & Tools):** ~9.5 lbs (~4.3 kg)
- *Solid, substantial heirloom feel that remains comfortable to carry via the rounded 1" walnut center handle!*

🔪 Joinery Guide & Technical Diagrams

 Primary Corner Joinery Option A: Box Joints (Finger Joints) — [Recommended]

- **Description:** Interlocking square fingers (1/4" width). Offers maximum glue surface area and high visual appeal on resawn walnut.
- **How to Cut:** Use a 1/4" dado blade on the table saw with a simple box joint pin jig, or a router table with a 1/4" straight bit.
- **Setup:** Fingers are cut 1/4" wide × 3/8" deep on 3/8" tray sides (and 1/4" wide × 1/2" deep on 1/2" base box sides).



 Primary Corner Joinery Option B: Mitered Corners with White Ash Splines — [Easiest & Decorative]

- **Description:** 45° mitered corners reinforced with 1/8" thick White Ash splines cut across the outer corner.
- **How to Cut:** Miter tray walls at 45°. Glue up box with strap clamps. Once cured, run outer corners over a table saw blade set to 45° in a spline jig. Glue 1/8" White Ash key splines into kerfs and trim flush!

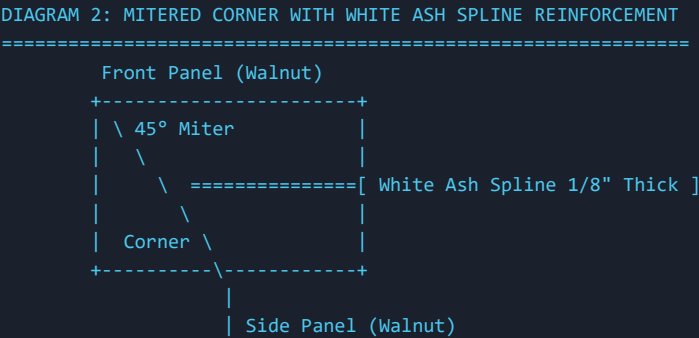


 Table Saw 45° Octagon Technique for Stationary Handle — [\$0 Lathe-Free Rounding]

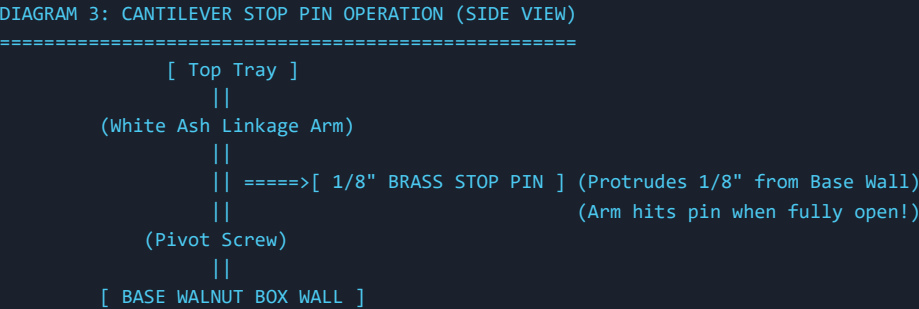
- **Method:** Rip 1" × 1" square walnut handle blank. Tilt table saw blade to 45°. Pass all 4 long corners over blade to create an 8-sided octagon. Finish with 100 & 220 grit sandpaper using a "shoe-shine" back-and-forth motion to roll corners into a perfectly smooth 1" round grip dowel!

 Cantilever Arm Stop Pins (Over-Extension Protection)

- **Purpose:** Stops the White Ash arms at a clean 90°/105° angle when opening so the trays stay

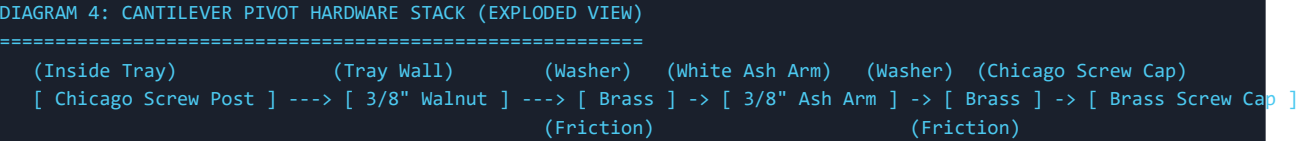
rock-solid and don't tip backwards.

- **Hardware:** 1/8" Solid Brass Rod / Dowel Pin (or 1/8" White Ash wooden pin). Press-fit into a 1/8" hole drilled 1/4" deep in the base box wall, leaving 1/8" protruding.



 Embedded Neodymium Lid Magnets (Invisible Closure)

- **Method:** Drill 1/4" dia × 1/16" deep holes in the top edges of upper trays and underside of lids. Press N52 Neodymium disk magnets flush using epoxy (match polarities!). Yields a clean, latch-free "snap" closure.



 Final Dimensions Summary

Dimension	Exterior (Nominal)	Interior Tray Dimensions	Notes
Length (L)	16.0" (406 mm)	14.75" (375 mm)	Fits two 7.25" pencils end-to-end OR rows of 30+ pencils
Depth (D)	8.5" (216 mm)	7.5" (190 mm)	Fits standard 7.25" Prismacolor/Faber pencils lying across
Height (Closed)	7.5" (190 mm)	Base: 3.25" / Trays: 1.25" ea	Closed box height (excluding handle)
Height (Total)	12.5" (318 mm)	Handle clearance: 5.0"	Stationary center handle clearance for hand grip

 Cutting List (Rough-Cut Lumber Milling Targets)

Component	Qty	Finished Length	Width	Thickness	Material & Notes
Base Box Front & Back	2	16.0"	3.5"	1/2" (0.50")	Solid Black Walnut

Base Box Ends	2	8.5"	3.5"	1/2" (0.50")	Solid Black Walnut
Base Box Bottom	1	15.0"	7.5"	1/4" (0.25")	Resawn Walnut or Plywood
Stationary Handle Post	2	12.5"	1.25"	5/8" (0.625")	Solid Black Walnut
Stationary Handle Grip	1	9.0"	1.0"	1.0" round	Rounded Black Walnut dowel (Table saw octagon method)
Middle Trays Front/Back	4	7.75"	1.5"	3/8" (0.375")	Solid Black Walnut (2 trays: left & right)
Middle Trays Sides	4	7.5"	1.5"	3/8" (0.375")	Solid Black Walnut
Middle Trays Bottom	2	7.0"	7.0"	1/4" (0.25")	Resawn Walnut (Velvet lined)
Top Trays Front/Back	4	7.75"	1.5"	3/8" (0.375")	Solid Black Walnut (2 lidded top trays)
Top Trays Sides	4	7.5"	1.5"	3/8" (0.375")	Solid Black Walnut
Top Trays Bottom	2	7.0"	7.0"	1/4" (0.25")	Resawn Walnut
Top Lids	2	8.0"	7.75"	3/8" (0.375")	Bookmatched Black Walnut
Cantilever Linkage Arms	8	5.5"	0.75"	3/8" (0.375")	Rough-Cut White Ash (High contrast accent!)

Step-by-Step Milling, Assembly & Finishing Guide

- 1. Milling Rough Stock:** Surface-plane rough Walnut to 1/2", 3/8", and 1/4". Plane rough White Ash offcuts to exact 3/8" thickness and rip to 3/4" width for the 8 linkage arms.
- 2. Cantilever Arm Precision Drilling:** Precision-drill 3/16" holes in the 8 White Ash arms at exact **4.0" center-to-center hole spacing** on the drill press. Chamfer/round arm ends for smooth movement.
- 3. Table Saw Handle Rounding:** Rip 1" × 1" walnut blank. Tilt table saw blade to 45° to cut an 8-sided octagon. Sand smooth using shoe-shine motion.
- 4. Corner Joinery & Tray Assembly:** Cut Box Joints (Option A) or Mitered Joints with White Ash splines (Option B). Rout 1/4" bottom grooves. Glue and clamp square.
- 5. Stop Pins & Magnets:** Drill 1/8" holes in base box side walls and press-fit 1/8" brass stop pins (leaving 1/8" sticking out). Mortise 1/4" × 1/16" N52 neodymium magnets into lid edges with epoxy.
- 6. Finishing Protocol:** Apply **Danish Oil** to tray interiors (zero-tack core protection). Apply 3-4 coats of **Pure Tung Oil** to exterior walnut and White Ash arms (UV resistance & hard curing). Buff exterior with paste wax.
- 7. Central Handle & Arm Installation:** Mortise handle posts into base box. Assemble White Ash arms to Walnut trays using brass Chicago screws.